



2004 Network Tariff Report

Produced pursuant to clause 11.3.5 of Volume I of the Office of the Regulator General's Electricity Distribution Price Determination 2001 – 2005, and supplementary advisory notes

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United Energy Tariff Report 2004

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1. Purpose

This report is intended to give guidance to customers, setting out the issues, principles and rationale used by United Energy in setting its Network Use of System (NUOS) tariffs for 2004. Consultation with customers and groups representing customers continued over 2003. However, United Energy welcomes further comments from interested parties and has demonstrated its willingness to incorporate such views into future tariff settings through some of the changes implemented over the past three years.

This 2004 tariff report can be read in conjunction with the previous two tariff reports, available on the Essential Services Commission (ESC) (www.esc.vic.gov.au) and United Energy (www.unitedenergy.com.au) websites.

2. Tariff setting principles (basis for tariff policies)

The design of the tariffs sought to incorporate the following policy principles:

- Regulatory compliance. Compliance with the regulatory requirements of the Office of the Regulator General (ORG) and Essential Services Commission (ESC), as specified in and interpreted from Volumes I & II of the Electricity Distribution Price Determination, and subsequent partial re-determination. These regulatory requirements limit the extent to which tariffs can change, particularly any average price increases.
- Customer choice. To provide customers with meaningful choices of tariff options based upon behavioural response and shared benefits.
- Market equity. Having regard to previous price levels and hence relative changes. Further, pricing should apply to all retailers in a neutral manner and not impede the viability of full retail contestability.
- Cost reflectivity. Seek appropriate mechanisms for achieving inter-customer group equity in the allocation of distribution cost-recovery requirements. Pricing to recognise cost-reflectivity within the constraints imposed by regulatory requirements (ie. that tariffs fall within economic upper and lower bounds) and ensuring efficient investment signalling and network usage.
- Behavioural elasticity. Seek to utilise rational consumer behavioural elasticity, both in terms of usage responses to pricing signals and tariff switching.
- Practicality. Seek to simplify mechanisms balancing against economic functionality, while having regard to minimising transaction and pricing administration cost.
- Environmental. Within the limitations of the scope and context of electricity distribution pricing, have regard to opportunities to improve utilisation efficiency and accommodate emerging energy technologies, particularly in respect of greenhouse gas emission reduction.

Understandably, not all the principles could be satisfied to their full extent, while fully complying with the ESC's requirements, particularly as some pull in opposing directions. United Energy has sought to satisfy these principles to the greatest extent possible whilst maintaining the given regulatory constraints.

3. Changes in tariffs, tariff structure and new tariffs

The 2004 Network Use of System (NUOS) tariffs reflect the underlying structure of both the Transmission Use of System (TUOS) and Distribution Use of System (DUOS) charges. That is, the structures of the TUOS and DUOS tariffs are identical and the NUOS rates are the simple addition of DUOS and TUOS.

The fundamental restructuring of United Energy's tariffs was undertaken with the 2001 tariff set. The 2002 tariff submission contained some minor restructuring including the replacement of the interruptable tariff introduced in 2001 and two new tariffs for small customers. The 2004 tariff set, as per the 2003 tariff set contained no new tariffs.

4. Uptake of Time of Use (TOU) tariffs

In 2001 and 2002, United Energy introduced a number of optional Time of Use (TOU) tariffs for customers for use in conjunction with interval (or smart) meters. At the same time, several tariffs incorporating the historical 11pm to 7am Off Peak time zone, relying on obsolete metering technology, were closed to new connections. New connections requiring off peak network tariffs are now required to install an interval meter. It was intended that such a change in connection characteristics would trigger a change in the underlying network tariff to be billed to the customer's Retailer.

Delays with the ESC's formal position paper on the roll out of interval metering, have meant that United Energy's uptake of TOU tariffs has been substantially lower than forecast at the time of introduction of the 2001 tariffs. A further related issue is the reluctance of Retailers to repackage United Energy's TOU tariffs to smaller customers. To date, no Retailer has publicly made an offer based on the low volume TOU tariff. Informal discussions have revealed a number of operational reasons associated with these decisions, including that most customers currently would need to pay additional fees to have an interval meter installed to take advantage of the new tariffs.

At this point in time, the ESC is proposing an interval meter rollout to >160 MWh/annum customers over 2 years and to direct connect 3 and 2 phase plus electric hot water customers over 5 years. This is not yet finalised, and at this stage of the discussions, the ESC is still not allowing customers to be automatically reassigned to TOU tariffs following an interval meter changeover. As far as United Energy is concerned, the Interval Meter Rollout program is a moot point if customers cannot be automatically reassigned to more cost reflective TOU tariffs so as to result in efficient outcomes, ultimately delivering lower network prices for all customers. We encourage all interested parties to contribute to the ESC consultation process.

5. Distribution Use of System (DUOS) tariffs

The regulated price control formula allowed United Energy an average increase in the price control for distribution tariffs in 2004 of 0.165% over the 2003 tariffs. The major components of this increase were “CPI – X” (1.573%), and “S” (-1.387%). The term “CPI-X” is the underlying price path for distribution tariffs over the regulatory period (2001-2005). The “S” factor relates to network reliability and reflects the network reliability statistics achieved by United Energy’s management of its network assets.

Within the average tariff increase, the price control formulae allows individual tariffs to increase by a maximum of 4.651%, although the average increase across all customer classes is limited to 0.165%. This allows United Energy some flexibility as to where the average 0.165% DUOS price control increase is incurred.

All closed tariffs were increased by the maximum allowable price increase of 4.651%, which allowed the open tariffs (notably the TOU tariffs) to receive a 0.5% reduction in DUOS price to meet the average 0.165% price increase overall. As previously explained, United Energy’s TOU tariff uptake has been below expectations, and this effective DUOS price decrease is an added incentive and reward to those customers who wish to take up a TOU tariff. Additionally, the LV large KVA tariff received a slightly greater DUOS price reduction to make it more cost reflective as compared to the HV tariff.

Closed tariffs (non TOU) were increased by the maximum allowable to reflect their increasing contribution to co-incident peak demand due to the growth in cooling load amongst small and medium customers and the inability to distinguish between “dumb” profiles. This treats them all as if cooling load exists at the premise and customers in these classes without cooling load or who are prepared to respond to TOU price signals are incentivised to switch to one of the optional TOU tariffs available.

6. Transmission Use of System (TUOS) tariffs

The Grid Fees paid by United Energy do not perfectly reflect the underlying structure of the TUOS tariffs. This is an historical anomaly that has been maintained in order to retain the additive nature of structure of the TUOS and DUOS charges. Restructuring the TUOS tariffs to better reflect the underlying charges they are designed to recover would significantly increase the complexity of the NUOS charges faced by customers. No Victorian DB currently perfectly reflects the underlying structure of the Grid Fees charges by SPI Powernet and VENCORP. United Energy has however restricted the application of TUOS rates to those components of the NUOS which best reflect the underlying Grid Fees, (that is Peak Energy, Summer Demand Incentive Charge and Rolling Demand) and so Off Peak energy does not attract TUOS.

As forecast in 2003, the grid fees have increased again this year, this time by 10.5%. The TUOS rates should in fact have increased by 49.6% to recover the 10.5% grid fee rise and the shortfall United Energy received in TUOS revenue from 2003 (as in 2003, the TUOS price rise was 24.2% when it should have been 54% to recover all outstanding Transmission related costs). However, as in 2003, to avoid price spikes to the customer, the Regulator has only allowed United Energy to raise the TUOS rates by 23.1%. This decision means the United Energy carries a further cashflow cost of \$10M in unfunded grid fees for 2004. This exposes United Energy to significant regulatory risk in terms of cashflow and distorts TUOS passthrough pricing signals to the market. On a positive note, as the grid fees are not forecast to rise significantly in 2005, United Energy should be able to recover this lost revenue via the 2005 TUOS tariffs.

As a consequence of the above, all 2004 transmission tariff components were increased uniformly by the maximum allowed rate of 23.1%.

7. Network Use of System (NUOS) tariffs

NUOS charges are those seen on customer invoices and are the sum of Transmission and Distribution charges. Different customer classes have different contributions to overall Transmission requirements and have different TUOS contributions to their final NUOS charges.

Two of the tariffs, which are closed to new connections, are paying above their upper bound or “Stand-alone” cost. The LVL1R and LVL2R rates are historical tariffs for “Large” customers that do not incur a demand charge. These tariffs will continue to increase at the maximum rate until the regulatory price control mechanism allows these tariffs to be collapsed into the appropriate LV Medium tariff category or no customers remain on the tariff. Unfortunately the regulatory price control mechanism does not allow for compulsory transfer of customers from these obsolete tariffs.

All of the other tariffs sit between the “Avoidable” and “Stand-alone” costs for their tariff class.

8. Future direction of United Energy's tariffs

As discussed in the 2003 tariff report, United Energy is committed to seeking beneficial customer outcomes by providing choice and incentives via innovation in its distribution tariffs. Most of the planned initiatives have now been fully implemented and further changes are not planned in the short term, particularly prior to the next regulatory price reset.

There is, however, a watching brief on the following issues which may be taken into account in the next annual tariff setting exercise.

- Encouraging an increased uptake of interval-metering based tariffs such as the current 'TOU' series;
- Encouraging customer demand-side initiatives
- Preserving as far as possible a stable commercial climate for investment in DSM equipment or practices
- The SDIC concept will remain, but the time window may be updated from time to time in order that it remain aligned with the key network peak demands
- Cost-of-supply modelling updated to reflect changes in relative contributions from segments
- Further accentuating the emphasis on peak season (summer), day of week and time of day in order to stimulate DSM response
- Properly integrate the contribution that distribution-connected generator customers should be making to the costs of providing network services that all users share and derive value from.
- Further closure of tariffs based on obsolete metering.

9. Let us know your views

United Energy continues to encourage customers to discuss ideas on how mutual needs may be better met. Many of the 2004 changes will be familiar to those customers that participated in sessions hosted by UE during 2003. It is hoped that this can be further enhanced during the next year.

10. Appendix: 2004 NUOS, DUOS and TUOS tariff rates

See the ESC website at:

http://www.esc.vic.gov.au/apps/page/user/pdf/UE2004_Tariffs_2004.pdf

Or see the United Energy website at:

http://www.ue.com.au/industry/network_tariffs/download/internetsitetariffs_2004.pdf

11. Appendix: Glossary of terms

DSM Demand-side management. Refers to the ability of customers that consume a product to change behaviours that alter the overall balance between supply and demand. Examples may include the installation of energy management systems, or deferring consumption at time of peak demand.

ESC Essential Services Commission (Previously Office of the Regulator-General)

Load Duration Curve (LDC) A graphical representation of the utilisation of a network. For the time period under observation, the load on the network is shown, sorted by highest to lowest. The resulting area graphically presents the percentage of the available capacity utilised.

Load Factor Load factor is a measure of how 'steady' a load is. It is the extent to which average usage compares to maximum usage. It is generally much more efficient to supply a high load factor (100%=constant unchanging demand) than a 'peaky' (less than 100%) usage pattern. This is because depending upon the time of the peak, the capacity needs to be there all the time to supply the peak, even though most of the time it may not be fully required.

LVS1R etc Abbreviated Tariff names.

- The 1st two letters refer to the voltage level at which the customer is supplied, LV = low voltage 240V(single phase) or 415V(three phase), HV = high voltage 6.6, 11 or 22kV, ST = sub-transmission 22 and 66kV. HOT = tariff variant based on high ambient temperature threshold
- The 2nd section refers to the type of customer, S = small, M = medium, L = large, no letters generally indicates 'generic' interval metered customer.
- 3rd section indicates type of rate structure, 1R = single energy rate, 2R = peak/off-peak rate, kW = interval metered in kW, kVA = interval metered in kVA.

kW Kilowatts, (1,000 Watts) the ISO unit of power measurement

kWh An I.S.O. unit used for the measurement of energy. It is the product of kW times time (in hours).

kVA kilo-Volt-Ampere, the vector product of voltage times current. It is a measure of the proportion of useful (kW) power delivered vs. the current (Amps) drawn by the load, this is known as 'power factor'. The size of assets required is generally a function of the current. Therefore the closer the power factor is to 1.00, the smaller the asset required to supply the load.

k-factor mechanism A regulatory method aimed at ensuring differences between 'actual' and 'forecasted' are washed-up year to year on a revenue neutral basis. This applies to Transmission wholesale costs and TUOS revenues.

Residue settlement A means by which over-recoveries of costs on the part of the energy and transmission markets are rebated back to customers.

SDIC Summer Demand Incentive Charge